

Appendix of On the Geometric Interpretation of Non-Singular Black Hole Interiors

Appendix-A.

Let us begin with the tetrad given in spherical coordinates,

$$(e^{(0)}, e^{(1)}, e^{(2)}, e^{(3)}) = (dt, dr, r d\theta, r \sin(\theta) d\varphi). \quad (\text{A1})$$

The transformation to the tetrad given in Eq. (??) is achieved by the standard spatial rotation, represented by an SO(3) matrix:

$$R(\theta, \varphi) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \sin(\theta) \cos(\varphi) & \cos(\theta) \cos(\varphi) & -\sin(\varphi) \\ 0 & \sin(\theta) \sin(\varphi) & \cos(\theta) \sin(\varphi) & \cos(\varphi) \\ 0 & \cos(\theta) & -\sin(\theta) & 0 \end{pmatrix}. \quad (\text{A2})$$

The rotated tetrad is obtained via

$$e'^{(\mu)} = R^{(\mu)}_{(\nu)} e^{(\nu)}. \quad (\text{A3})$$

Explicitly, one finds

$$e'^{(0)} = dt, \quad (\text{A4})$$

$$e'^{(1)} = \sin(\theta) \cos(\varphi) dr + r \cos(\theta) \cos(\varphi) d\theta - r \sin(\theta) \sin(\varphi) d\varphi, \quad (\text{A5})$$

$$e'^{(2)} = \sin(\theta) \sin(\varphi) dr + r \cos(\theta) \sin(\varphi) d\theta + r \sin(\theta) \cos(\varphi) d\varphi, \quad (\text{A6})$$

$$e'^{(3)} = \cos(\theta) dr - r \sin(\theta) d\theta. \quad (\text{A7})$$

Finally, implementing the transformation

$$dr \longrightarrow dr + W(r) dt, \quad (\text{A8})$$

yields the tetrad components

$$e'^{(a)}{}_{\mu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ W(r) \sin(\theta) \cos(\varphi) & \sin(\theta) \cos(\varphi) & r \cos(\theta) \cos(\varphi) & -r \sin(\theta) \sin(\varphi) \\ W(r) \sin(\theta) \sin(\varphi) & \sin(\theta) \sin(\varphi) & r \cos(\theta) \sin(\varphi) & r \sin(\theta) \cos(\varphi) \\ W(r) \cos(\theta) & \cos(\theta) & -r \sin(\theta) & 0 \end{pmatrix}. \quad (\text{A9})$$

Appendix-B. Velocity-Gradient Invariants and the Nonlinear GDV Lagrangian

The first subsection of this Appendix is dedicated to show the relation between tensor invariants and velocity gradient. The next subsection continues by non-linear Lagrangian choice from these introduced invariants.

Velocity Gradient Tensor:

A vector field with a pure spherical symmetry can be defined by:

$$\mathbf{W}(\mathbf{r}) = \zeta(r) \hat{\mathbf{r}} \quad (\text{A10})$$

Divergence of $\mathbf{W}$ in spherical coordinates is calculated as below

$$\nabla \cdot \mathbf{W} = \frac{1}{r^2} \frac{d}{dr} [r^2 \zeta(r)] = \zeta'(r) + \frac{2}{r} \zeta(r) \quad (\text{A11})$$

Let us define velocity gradient tensor, $\mathcal{L}_{ij} = \nabla_i W_j$, in spherical coordinates

$$\mathcal{L} = \begin{pmatrix} \frac{\partial \zeta_r}{\partial r} & \frac{1}{r} \left(\frac{\partial \zeta_r}{\partial \theta} - \zeta_\theta \right) & \frac{1}{r \sin \theta} \frac{\partial \zeta_r}{\partial \phi} - \frac{\zeta_\phi}{r} \\ \frac{\partial \zeta_\theta}{\partial r} & \frac{1}{r} \left(\frac{\partial \zeta_\theta}{\partial \theta} + \zeta_r \right) & \frac{1}{r \sin \theta} \frac{\partial \zeta_\theta}{\partial \phi} - \frac{\zeta_\phi \cot \theta}{r} \\ \frac{\partial \zeta_\phi}{\partial r} & \frac{1}{r} \frac{\partial \zeta_\phi}{\partial \theta} & \frac{1}{r \sin \theta} \frac{\partial \zeta_\phi}{\partial \phi} + \frac{\zeta_r}{r} + \frac{\zeta_\theta \cot \theta}{r} \end{pmatrix}. \quad (\text{A12})$$

Under angular independent motion directly reduces $\mathcal{L}$ to a diagonal form $\text{diag}(\zeta', \zeta/r, \zeta/r)$ that reveals eigenvalues of

$$\lambda_r = \zeta'(r), \quad \lambda_\theta = \frac{\zeta(r)}{r}, \quad \lambda_\phi = \frac{\zeta(r)}{r}. \quad (\text{A13})$$

Thus, divergence is equivalent of trace of velocity gradient tensor under spherical symmetry.

$$\nabla \cdot \mathbf{W} = \lambda_r + \lambda_\theta + \lambda_\phi \quad (\text{A14})$$

Principal invariants are given below for any rank-2 tensor:

$$I_1 = \text{tr}(\mathcal{L}), \quad I_2 = \frac{1}{2} [\text{tr}(\mathcal{L})^2 - \text{tr}(\mathcal{L}^2)], \quad I_3 = \det(\mathcal{L}) \quad (\text{A15})$$

In addition and velocity gradient tensor can be divided into two parts:

$$\mathcal{L} = \{ \mathcal{L} - (1/3) \text{tr}(\mathcal{L}) I \} + \{ \text{tr}(\mathcal{L}) I \} \quad (\text{A16})$$

The first term on the r.h.s is traceless and called "deviatoric tensor". It carries the info of distortion and rotation. On the other hand, the second term is called "isotropic" and demonstrates the rate of expansion / contraction. Under the pure radial symmetry, $\mathcal{L}$ is identically isotropic.

Finally, definitions of main invariants can be given by:

$$J_i = \sum_j (\lambda_j)^i \quad (\text{A17})$$

and therefore J_2 gives total energy density of local deformations.

Nonlinear Lagrangian:

For observer-independent formulation, the most general Lagrangian candidate can be written in terms of the principal invariants of the velocity-gradient tensor as

$$L = \frac{e}{2\kappa} f(I_1, I_2, I_3). \quad (\text{A18})$$

In the weak-field limit, consistency with the teleparallel equivalent of general relativity requires the leading-order term to reproduce the torsion scalar T . As shown in Section 5, for the drift-tetrad structure considered in this work, the torsion scalar is directly related to the second principal invariant as $T = 2I_2$. Therefore, the leading term in the gravitational Lagrangian must be $2I_2$. Apart from this requirement, a term linear in I_1 does not represent an independent bulk contribution. Since $I_1 = \nabla \cdot W$, its linear contribution corresponds to a total divergence and can only generate a boundary term in the action. The central physical motivation of the present work is the inclusion of nonlinear strong-field corrections. Since $T = 2I_2$, the natural quadratic self-interaction of the torsion scalar is $T^2 \sim I_2^2$. This term is negligible in the weak-field regime but can become relevant in the high-gradient region near the black-hole core. Moreover, for the Schwarzschild exterior drift solution one has: $I_2 \rightarrow 0$. So any correction of the form $(I_2)^n$ automatically preserves the exterior vacuum solution. By contrast, terms involving $(I_1)^n$, $(I_3)^n$ or any other possible combination do not generally vanish for the Schwarzschild exterior and may therefore disturb the vacuum branch unless additional conditions or suppression mechanisms are imposed. Taking these points together, the minimal nonlinear Lagrangian that naturally preserves the exterior Schwarzschild vacuum while allowing a regular isotropic interior solution is

$$L_{\text{Total}} = \frac{e}{2\kappa} (2I_2 + \chi_1 I_2^2). \quad (\text{A19})$$

Higher order terms ($n > 2$) are mathematically possible but they are assumed to be negligible in general. Beyond this minimal choice, one may also construct geometrically and physically meaningful combinations from the traceless, or deviatoric, part of the velocity-gradient tensor. Let

$$\Sigma = \mathcal{L} - \frac{I_1}{3} \mathbf{1}, \quad (\text{A20})$$

where $\mathcal{L}$ denotes the velocity-gradient tensor. The corresponding deviatoric invariants may be written as

$$S_1 = \text{tr}(\Sigma) = 0, \quad (\text{A21})$$

$$S_2 = \text{tr}(\Sigma^2) = \frac{2}{3} (I_1^2 - 3I_2), \quad (\text{A22})$$

and

$$S_3 = \det(\Sigma) = I_3 - \frac{1}{3} I_1 I_2 + \frac{2}{27} I_1^3. \quad (\text{A23})$$

Here S_1 vanishes identically because the deviatoric tensor carries no volumetric trace. The invariant S_2 measures the magnitude of anisotropic deformation, while S_3 encodes the cubic character of the anisotropic distortion. Both S_2 and S_3 vanish for a fully isotropic velocity-gradient field. These terms are not required in the minimal static and spherically symmetric model. However, they become natural candidates in more general geometries. For example, in rotating or axially symmetric black-hole configurations, such as Kerr-type systems, the velocity-gradient tensor may contain shear, anisotropic deformation, and rotational couplings. In that case, terms involving S_2 , S_2^2 , or $I_1 S_3$ may provide physically meaningful corrections to the Lagrangian. Their inclusion would describe deviations from purely isotropic drift dynamics and could capture strong-field anisotropic effects that are absent in the minimal Schwarzschild-like construction. Nevertheless, necessary terms (including rotation or not) would already manifest themselves naturally in tensor scalar: T such that $L_{Total} = T + \chi_1 T^2$ keeps its form.

Appendix-C. Variational Derivation of the Spherically Symmetric Drift Field

In Appendix-B, it has been shown that velocity gradient tensor can be rewritten as below under pure radial drift

$$\mathcal{L} = \text{diag}\left(\frac{d}{dr}\zeta(r), \frac{\zeta(r)}{r}, \frac{\zeta(r)}{r}\right) \quad (\text{A24})$$

On the other hand, total Lagrangian is given by

$$L_{Total} = \frac{e}{2\kappa}(T + \chi_1 I_2^2). \quad (\text{A25})$$

where $e = r^2 \sin(\theta)$ in spherical coordinates and $T = 2I_2$ is satisfied in the current study. If the Euler-Lagrange equation for the total Lagrangian is examined in the $(\zeta, \zeta'; r)$ coordinates, then we obtain:

$$\frac{d}{dr} \frac{\partial L_{Total}}{\partial \zeta'} - \frac{\partial L_{Total}}{\partial \zeta} = 0. \quad (\text{A26})$$

Implementing Eq. (A25) into Eq. (A26) gives

$$\zeta(r)(\zeta''(r) - \frac{\zeta(r)}{r^2}) + \zeta'(r)^2 = 0 \quad (\text{A27})$$

which has a solution

$$W(r) = \zeta(r)\hat{r}, \quad \zeta(r) = \pm \frac{\sqrt{c_1 r^4 + c_2^2 r}}{r}. \quad (\text{A28})$$

Appendix-D. Lorentz Covariance Test for a Radial Shift Tetrad

Consider the tetrad

$$e^0 = dt, \quad (\text{A29})$$

$$e^i = a(t) dx^i + W(t, r) n_i dt, \quad (\text{A30})$$

where

$$r = \sqrt{x^k x_k}, \quad (\text{A31})$$

and

$$n_i = \frac{x_i}{r}. \quad (\text{A32})$$

Define

$$W_i = W(t, r) n_i. \quad (\text{A33})$$

We begin in the Weitzenböck gauge

$$\omega^a_b = 0. \quad (\text{A34})$$

The torsion 2-form is

$$T^a = de^a + \omega^a_b \wedge e^b. \quad (\text{A35})$$

Since initially

$$\omega^a_b = 0, \quad (\text{A36})$$

we have

$$T^a = de^a. \quad (\text{A37})$$

Initial Torsion

First compute

$$T^0. \quad (\text{A38})$$

Because

$$e^0 = dt, \quad (\text{A39})$$

we obtain

$$T^0 = de^0 = d(dt) = 0. \quad (\text{A40})$$

Therefore

$$\boxed{T^0 = 0} \quad (\text{A41})$$

Now compute the spatial torsion:

$$T^i = de^i. \quad (\text{A42})$$

Using

$$e^i = a(t) dx^i + W n_i dt, \quad (\text{A43})$$

we find

$$de^i = d(a dx^i) + d(W n_i dt). \quad (\text{A44})$$

Derivative of the FLRW Part

First compute

$$d(a \, dx^i). \quad (\text{A45})$$

Applying the exterior derivative,

$$d(a \, dx^i) = da \wedge dx^i. \quad (\text{A46})$$

Since

$$da = \dot{a} \, dt, \quad (\text{A47})$$

we obtain

$$d(a \, dx^i) = \dot{a} \, dt \wedge dx^i. \quad (\text{A48})$$

Derivative of the Radial Shift

Now compute

$$d(W n_i \, dt). \quad (\text{A49})$$

Using the Leibniz rule,

$$d(W n_i \, dt) = d(W n_i) \wedge dt + W n_i \, d(dt). \quad (\text{A50})$$

Since

$$d(dt) = 0, \quad (\text{A51})$$

this becomes

$$d(W n_i \, dt) = d(W n_i) \wedge dt. \quad (\text{A52})$$

Now expand

$$d(W n_i). \quad (\text{A53})$$

Because $W n_i$ depends on both t and x^j ,

$$d(W n_i) = \partial_j(W n_i) \, dx^j + \partial_t(W n_i) \, dt. \quad (\text{A54})$$

Therefore,

$$d(W n_i) \wedge dt = \partial_j(W n_i) \, dx^j \wedge dt + \partial_t(W n_i) \, dt \wedge dt. \quad (\text{A55})$$

Since

$$dt \wedge dt = 0, \quad (\text{A56})$$

we get

$$d(W n_i) \wedge dt = \partial_j(W n_i) \, dx^j \wedge dt. \quad (\text{A57})$$

Using antisymmetry,

$$dx^j \wedge dt = -dt \wedge dx^j, \quad (\text{A58})$$

we obtain

$$d(W n_i) \wedge dt = -\partial_j(W n_i) \, dt \wedge dx^j. \quad (\text{A59})$$

Substituting back,

$$T^i = \dot{a} dt \wedge dx^i - \partial_j(W n_i) dt \wedge dx^j. \quad (\text{A60})$$

Rewrite the first term:

$$\dot{a} dt \wedge dx^i = \dot{a} \delta^i_j dt \wedge dx^j. \quad (\text{A61})$$

Thus

$$T^i = [\dot{a} \delta^i_j - \partial_j(W n_i)] dt \wedge dx^j. \quad (\text{A62})$$

Computing $\partial_j(W n_i)$

Now compute

$$\partial_j(W n_i). \quad (\text{A63})$$

Applying the product rule,

$$\partial_j(W n_i) = (\partial_j W) n_i + W \partial_j n_i. \quad (\text{A64})$$

Since $W = W(t, r)$,

$$\partial_j W = W_r n_j, \quad (\text{A65})$$

where

$$W_r = \frac{\partial W}{\partial r}. \quad (\text{A66})$$

Now compute

$$\partial_j n_i. \quad (\text{A67})$$

Using

$$n_i = \frac{x_i}{r}, \quad (\text{A68})$$

we obtain

$$\partial_j n_i = \partial_j \left(\frac{x_i}{r} \right). \quad (\text{A69})$$

Applying the quotient rule,

$$\partial_j n_i = \frac{\delta_{ij}}{r} - \frac{x_i}{r^2} \partial_j r. \quad (\text{A70})$$

Since

$$\partial_j r = \frac{x_j}{r} = n_j, \quad (\text{A71})$$

and

$$x_i = r n_i, \quad (\text{A72})$$

we get

$$\partial_j n_i = \frac{\delta_{ij}}{r} - \frac{n_i n_j}{r}. \quad (\text{A73})$$

Therefore

$$\boxed{\partial_j n_i = \frac{1}{r} (\delta_{ij} - n_i n_j)} \quad (\text{A74})$$

Substituting into $\partial_j(W n_i)$,

$$\partial_j(W n_i) = W_r n_i n_j + \frac{W}{r} (\delta_{ij} - n_i n_j). \quad (\text{A75})$$

Initial Spatial Torsion

Substituting into T^i ,

$$T^i = \left[\dot{a} \delta^i_j - W_r n_i n_j - \frac{W}{r} (\delta_{ij} - n_i n_j) \right] dt \wedge dx^j. \quad (\text{A76})$$

Therefore

$$T^i = \left[\dot{a} \delta^i_j - W_r n_i n_j - \frac{W}{r} (\delta_{ij} - n_i n_j) \right] dt \wedge dx^j \quad (\text{A77})$$

The radial eigenvalue is

$$T_{\parallel} = \dot{a} - W_r \quad (\text{A78})$$

The transverse eigenvalues are

$$T_{\perp} = \dot{a} - \frac{W}{r} \quad (\text{A79})$$

Infinitesimal Radial Lorentz Boost

Now attempt to remove the radial shift by a local Lorentz boost.

The only infinitesimal boost parameter is selected as β^i and at linear order, the transformed spatial tetrad becomes

$$e'^i \simeq a dx^i + (W^i + \beta^i) dt. \quad (\text{A80})$$

Therefore, eliminating the shift contribution requires

$$\beta^i = -W^i. \quad (\text{A81})$$

Hence, within the infinitesimal boost sector, this choice is uniquely fixed by the requirement that the spatial tetrad becomes diagonal. The transformed tetrad is

$$e'^a = \Lambda^a_b e^b. \quad (\text{A82})$$

Spatial Tetrad

First compute

$$e'^i. \quad (\text{A83})$$

Using

$$e'^i = e^i + \beta_i e^0, \quad (\text{A84})$$

we obtain

$$e'^i = a dx^i + W n_i dt - W n_i dt. \quad (\text{A85})$$

Hence

$$e'^i = a dx^i \quad (\text{A86})$$

The radial shift disappears from the spatial tetrad.

Temporal Tetrad

Now compute

$$e'^0. \quad (\text{A87})$$

Using

$$e'^0 = e^0 + \beta_i e^i, \quad (\text{A88})$$

we obtain

$$e'^0 = dt + \beta_i (a dx^i + W n_i dt). \quad (\text{A89})$$

Substituting

$$\beta_i = -W n_i, \quad (\text{A90})$$

gives

$$e'^0 = dt - a W n_i dx^i - W^2 n_i n_i dt. \quad (\text{A91})$$

Since

$$n_i n_i = 1, \quad (\text{A92})$$

we obtain

$$e'^0 = (1 - W^2) dt - a W n_i dx^i. \quad (\text{A93})$$

Keeping only linear order in W ,

$$e'^0 \simeq dt - a W n_i dx^i. \quad (\text{A94})$$

Using

$$n_i dx^i = dr, \quad (\text{A95})$$

we get

$$\boxed{e'^0 \simeq dt - a W dr} \quad (\text{A96})$$

Generated Spin Connection

Initially

$$\omega^a_b = 0. \quad (\text{A97})$$

After the Lorentz transformation,

$$\omega'^a_b = \Lambda^a_c d(\Lambda^{-1})^c_b. \quad (\text{A98})$$

At linear order,

$$\omega'^i_0 = -d\beta_i. \quad (\text{A99})$$

Since

$$\beta_i = -W n_i, \quad (\text{A100})$$

we obtain

$$\boxed{\omega'^i_0 = d(W n_i)} \quad (\text{A101})$$

Now compute $d(W n_i)$.

Using the product rule,

$$d(Wn_i) = dW n_i + W dn_i. \quad (\text{A102})$$

Because $W = W(t, r)$,

$$dW = \dot{W} dt + W_r dr. \quad (\text{A103})$$

Using

$$dr = n_j dx^j, \quad (\text{A104})$$

we obtain

$$dW = \dot{W} dt + W_r n_j dx^j. \quad (\text{A105})$$

Now compute

$$dn_i. \quad (\text{A106})$$

Using

$$dn_i = \partial_j n_i dx^j, \quad (\text{A107})$$

and

$$\partial_j n_i = \frac{1}{r}(\delta_{ij} - n_i n_j), \quad (\text{A108})$$

we obtain

$$dn_i = \frac{1}{r}(\delta_{ij} - n_i n_j) dx^j. \quad (\text{A109})$$

Substituting everything back,

$$d(Wn_i) = n_i \dot{W} dt + W_r n_i n_j dx^j + \frac{W}{r}(\delta_{ij} - n_i n_j) dx^j. \quad (\text{A110})$$

Therefore

$\omega'^i{}_0 = n_i \dot{W} dt + W_r n_i n_j dx^j + \frac{W}{r}(\delta_{ij} - n_i n_j) dx^j$

(\text{A111})

Transformed Spatial Torsion

Now compute

$$T'^i = de'^i + \omega'^i{}_0 \wedge e'^0. \quad (\text{A112})$$

Derivative Term

Since

$$e'^i = a dx^i, \quad (\text{A113})$$

we obtain

$$de'^i = d(a dx^i) = da \wedge dx^i. \quad (\text{A114})$$

Using

$$da = \dot{a} dt, \quad (\text{A115})$$

we get

$$de'^i = \dot{a} dt \wedge dx^i. \quad (\text{A116})$$

Connection Term

Now compute

$$\omega'^i{}_0 \wedge e'^0. \quad (\text{A117})$$

Using

$$e'^0 = dt - aW dr, \quad (\text{A118})$$

we obtain

$$\omega'^i{}_0 \wedge e'^0 = d(Wn_i) \wedge (dt - aW dr). \quad (\text{A119})$$

Expanding,

$$\omega'^i{}_0 \wedge e'^0 = d(Wn_i) \wedge dt - aW d(Wn_i) \wedge dr. \quad (\text{A120})$$

The second term is quadratic in W :

$$W d(Wn_i) \sim W^2. \quad (\text{A121})$$

At linear order,

$$\omega'^i{}_0 \wedge e'^0 \simeq d(Wn_i) \wedge dt. \quad (\text{A122})$$

Now substitute

$$d(Wn_i) = \partial_j(Wn_i) dx^j + \partial_t(Wn_i) dt. \quad (\text{A123})$$

Then

$$d(Wn_i) \wedge dt = \partial_j(Wn_i) dx^j \wedge dt. \quad (\text{A124})$$

Using antisymmetry,

$$dx^j \wedge dt = -dt \wedge dx^j, \quad (\text{A125})$$

we obtain

$$d(Wn_i) \wedge dt = -\partial_j(Wn_i) dt \wedge dx^j. \quad (\text{A126})$$

Result for T'^i

Therefore

$$T'^i = \dot{a} dt \wedge dx^i - \partial_j(Wn_i) dt \wedge dx^j. \quad (\text{A127})$$

Thus

$$T'^i = \left[\dot{a} \delta^i{}_j - W_r n_i n_j - \frac{W}{r} (\delta_{ij} - n_i n_j) \right] dt \wedge dx^j. \quad (\text{A128})$$

Hence

$$\boxed{T'^i = T^i} \quad (\text{A129})$$

at linear order.

Transformed Temporal Torsion

Now compute

$$T'^0 = de'^0 + \omega'^0{}_i \wedge e'^i. \quad (\text{A130})$$

Derivative of e'^0

Using

$$e'^0 = dt - aWn_i dx^i, \quad (\text{A131})$$

we obtain

$$de'^0 = -d(aWn_i) \wedge dx^i. \quad (\text{A132})$$

Now compute

$$d(aWn_i). \quad (\text{A133})$$

Applying the product rule,

$$d(aWn_i) = da Wn_i + a d(Wn_i). \quad (\text{A134})$$

Using

$$da = \dot{a} dt, \quad (\text{A135})$$

we obtain

$$d(aWn_i) = \dot{a}Wn_i dt + a d(Wn_i). \quad (\text{A136})$$

Therefore

$$de'^0 = -\dot{a}Wn_i dt \wedge dx^i - a d(Wn_i) \wedge dx^i. \quad (\text{A137})$$

Connection Contribution

Now compute

$$\omega'^0{}_i \wedge e'^i. \quad (\text{A138})$$

Using

$$\omega'^0{}_i = d(Wn_i), \quad (\text{A139})$$

and

$$e'^i = a dx^i, \quad (\text{A140})$$

we obtain

$$\omega'^0{}_i \wedge e'^i = a d(Wn_i) \wedge dx^i. \quad (\text{A141})$$

Result for T'^0

Adding both contributions,

$$T'^0 = -\dot{a}Wn_i dt \wedge dx^i - a d(Wn_i) \wedge dx^i + a d(Wn_i) \wedge dx^i. \quad (\text{A142})$$

The spatial two-form terms cancel:

$$-a d(Wn_i) \wedge dx^i + a d(Wn_i) \wedge dx^i = 0. \quad (\text{A143})$$

Hence

$$T'^0 = -\dot{a}Wn_i dt \wedge dx^i. \quad (\text{A144})$$

Using

$$n_i dx^i = dr, \quad (\text{A145})$$

we finally obtain

$$\boxed{T'^0 = -\dot{a}W dt \wedge dr} \quad (\text{A146})$$

Initially,

$$\boxed{T^0 = 0} \quad (\text{A147})$$

Therefore,

$$T'^0 \neq T^0. \quad (\text{A148})$$

In general, by combining T^0 and T^i ,

$$T'^a \neq T^a. \quad (\text{A149})$$

Appendix-E. Preservation of torsion invariants under a radial Lorentz boost

When the spin connection is included, the torsion tensor transforms covariantly under a local Lorentz transformation:

$$T'^a{}_{bc} = \Lambda^a{}_d (\Lambda^{-1})^e{}_b (\Lambda^{-1})^f{}_c T^d{}_{ef}.$$

For a radial boost,

$$\Lambda^a{}_b = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \gamma = \frac{1}{\sqrt{1-\beta^2}}.$$

Before the boost, the independent nonvanishing components of the torsion tensor are

$$T^i{}_{0i} = K_i, \quad T^i{}_{i0} = -K_i,$$

where

$$K_1 = K_{\parallel}, \quad K_2 = K_3 = K_{\perp}.$$

Therefore, from the definition

$$T_a = T^b{}_{ba},$$

the torsion vector becomes

$$T_0 = T^1{}_{10} + T^2{}_{20} + T^3{}_{30} = -K_{\parallel} - 2K_{\perp}, \quad T_i = 0.$$

Hence,

$$T_a = (-K_{\parallel} - 2K_{\perp}, 0, 0, 0).$$

After the boost, the torsion vector transforms as a covector:

$$T'_a = (\Lambda^{-1})^b{}_a T_b.$$

Thus,

$$\begin{aligned} T'_0 &= \gamma T_0 = -\gamma (K_{\parallel} + 2K_{\perp}), \\ T'_1 &= \gamma\beta T_0 = -\gamma\beta (K_{\parallel} + 2K_{\perp}), \end{aligned}$$

and

$$T'_2 = T'_3 = 0.$$

Therefore,

$$T'_a = (-\gamma(K_{\parallel} + 2K_{\perp}), -\gamma\beta(K_{\parallel} + 2K_{\perp}), 0, 0).$$

We now compute the norm of the torsion vector. Using the Minkowski metric

$$\eta_{ab} = \text{diag}(-1, 1, 1, 1),$$

we obtain

$$T'_a T'^a = -(T'_0)^2 + (T'_1)^2.$$

Substituting the boosted components,

$$T'_a T'^a = -\gamma^2 (K_{\parallel} + 2K_{\perp})^2 + \gamma^2 \beta^2 (K_{\parallel} + 2K_{\perp})^2.$$

Factoring out the common term gives

$$T'_a T'^a = -\gamma^2 (1 - \beta^2) (K_{\parallel} + 2K_{\perp})^2.$$

Since

$$\gamma^2 = \frac{1}{1 - \beta^2},$$

it follows that

$$\gamma^2 (1 - \beta^2) = 1.$$

Hence,

$$T'_a T'^a = -(K_{\parallel} + 2K_{\perp})^2.$$

This is identical to the pre-boost value:

$$T'_a T'^a = T_a T^a.$$

Similarly, consider the first torsion invariant

$$I_1 = T_{abc} T^{abc}.$$

Before the boost,

$$I_1 = -2K_{\parallel}^2 - 4K_{\perp}^2.$$

Since the spin connection is included, the torsion tensor transforms as a true tensor, and all Lorentz matrices cancel inside the contraction:

$$T'_{abc} T'^{abc} = T_{abc} T^{abc}.$$

Therefore,

$$I'_1 = -2K_{\parallel}^2 - 4K_{\perp}^2 = I_1.$$

The second torsion invariant,

$$I_2 = T_{abc}T^{cba},$$

behaves analogously. Before the boost,

$$I_2 = -K_{\parallel}^2 - 2K_{\perp}^2.$$

After the boost,

$$I'_2 = T'_{abc}T'^{cba} = T_{abc}T^{cba} = -K_{\parallel}^2 - 2K_{\perp}^2.$$

Hence,

$$I'_2 = I_2.$$

The axial torsion vector is defined by

$$A^a = \frac{1}{6}\epsilon^{abcd}T_{bcd}.$$

For the present geometry,

$$A^a = 0$$

before the boost. Since the zero vector remains zero under Lorentz transformations,

$$A'^a = 0.$$

Consequently,

$$A'_a A'^a = 0 = A_a A^a.$$

The TEGR torsion scalar is given by

$$T = \frac{1}{4}T_{abc}T^{abc} + \frac{1}{2}T_{abc}T^{cba} - T_a T^a.$$

Equivalently,

$$T = \frac{1}{4}I_1 + \frac{1}{2}I_2 - T_a T^a.$$

Substituting the pre-boost expressions,

$$T = \frac{1}{4}(-2K_{\parallel}^2 - 4K_{\perp}^2) + \frac{1}{2}(-K_{\parallel}^2 - 2K_{\perp}^2) + (K_{\parallel} + 2K_{\perp})^2.$$

The first two terms yield

$$-\frac{1}{2}K_{\parallel}^2 - K_{\perp}^2 - \frac{1}{2}K_{\parallel}^2 - K_{\perp}^2 = -K_{\parallel}^2 - 2K_{\perp}^2.$$

The last term becomes

$$(K_{\parallel} + 2K_{\perp})^2 = K_{\parallel}^2 + 4K_{\parallel}K_{\perp} + 4K_{\perp}^2.$$

Therefore,

$$T = -K_{\parallel}^2 - 2K_{\perp}^2 + K_{\parallel}^2 + 4K_{\parallel}K_{\perp} + 4K_{\perp}^2.$$

This simplifies to

$$T = 4K_{\parallel}K_{\perp} + 2K_{\perp}^2,$$

or equivalently,

$$T = 2K_{\perp} (2K_{\parallel} + K_{\perp}).$$

After the boost,

$$I'_1 = I_1, \quad I'_2 = I_2, \quad T'_a T'^a = T_a T^a,$$

and therefore

$$T' = \frac{1}{4}I'_1 + \frac{1}{2}I'_2 - T'_a T'^a = \frac{1}{4}I_1 + \frac{1}{2}I_2 - T_a T^a.$$

Hence,

$$T' = T.$$

Explicitly,

$$T' = 2K_{\perp} (2K_{\parallel} + K_{\perp}).$$

Finally, under a radial Lorentz boost with the spin connection included,

$$I'_1 - I_1 = 0, \quad I'_2 - I_2 = 0, \quad T'_a T'^a - T_a T^a = 0,$$

$$A'_a A'^a - A_a A^a = 0, \quad T' - T = 0.$$

Therefore, all torsion invariants remain unchanged after the boost. This explicitly demonstrates the local Lorentz covariance of the formalism and shows that the boost does not modify the physical scalar quantities.

Appendix-F. Torsion Scalar Analysis of the Interior and Exterior Regions of Black Holes

In this appendix we explicitly derive the torsion scalar associated with the shifted tetrad used in the main text. We work in the Weitzenböck gauge, where the spin connection is taken to vanish.

The tetrad is

$$e^0 = dt, \quad e^i = a(t) dx^i + \epsilon W_i(t, \mathbf{x}) dt, \quad (\text{A150})$$

The Weitzenböck connection is defined by

$$\Gamma^\rho_{\mu\nu} = e_a{}^\rho \partial_\nu e^a{}_\mu. \quad (\text{A151})$$

Since

$$e^i{}_0 = \epsilon W_i, \quad e^i{}_j = a(t) \delta^i_j, \quad (\text{A152})$$

we obtain

$$\partial_j e^i{}_0 = \epsilon \partial_j W_i, \quad (\text{A153})$$

and

$$\partial_0 e^i{}_j = \dot{a} \delta^i{}_j. \quad (\text{A154})$$

Therefore the relevant connection components are

$$\Gamma^i{}_{j0} = e_k{}^i \partial_0 e^k{}_j = \frac{1}{a} \dot{a} \delta^i{}_j = H \delta^i{}_j, \quad (\text{A155})$$

and

$$\Gamma^i{}_{0j} = e_k{}^i \partial_j e^k{}_0 = \frac{\epsilon}{a} \partial_j W_i. \quad (\text{A156})$$

Here

$$H = \frac{\dot{a}}{a}. \quad (\text{A157})$$

Torsion tensor

The torsion tensor is

$$T^\rho{}_{\mu\nu} = \Gamma^\rho{}_{\nu\mu} - \Gamma^\rho{}_{\mu\nu}. \quad (\text{A158})$$

Thus,

$$T^i{}_{0j} = \Gamma^i{}_{j0} - \Gamma^i{}_{0j}, \quad (\text{A159})$$

which gives

$$T^i{}_{0j} = H \delta^i{}_j - \frac{\epsilon}{a} \partial_j W_i. \quad (\text{A160})$$

Equivalently,

$$T^i{}_{j0} = -H \delta^i{}_j + \frac{\epsilon}{a} \partial_j W_i. \quad (\text{A161})$$

It is useful to define

$$A_{ij} \equiv T^i{}_{0j} = H \delta_{ij} - \frac{\epsilon}{a} \partial_j W_i. \quad (\text{A162})$$

The trace is therefore

$$\text{Tr}(A) = 3H - \frac{\epsilon}{a} (\partial_x W_1 + \partial_y W_2 + \partial_z W_3). \quad (\text{A163})$$

Equivalently,

$$\text{Tr}(A) = 3H - \frac{\epsilon}{a} \partial_i W_i. \quad (\text{A164})$$

Torsion scalar

We use the standard teleparallel torsion scalar

$$T = \frac{1}{4} T_{\rho\mu\nu} T^{\rho\mu\nu} + \frac{1}{2} T_{\rho\mu\nu} T^{\nu\mu\rho} - T_\rho T^\rho, \quad (\text{A165})$$

where

$$T_\mu = T^\nu{}_{\nu\mu}. \quad (\text{A166})$$

For the present tetrad, the torsion scalar may be written in terms of the matrix A_{ij} as

$$T = 6H^2 - \frac{4\epsilon H}{a} \partial_i W_i + \frac{\epsilon^2}{a^2} \left[(\partial_i W_i)^2 - \frac{1}{2} \partial_j W_i \partial_i W_j - \frac{1}{2} \partial_j W_i \partial_j W_i \right]. \quad (\text{A167})$$

Thus, to linear order in ϵ ,

$$T = 6H^2 - \frac{4\epsilon H}{a} \partial_i W_i + \mathcal{O}(\epsilon^2). \quad (\text{A168})$$

Using $H = \dot{a}/a$, this becomes

$$T = 6 \left(\frac{\dot{a}}{a} \right)^2 - \frac{4\epsilon \dot{a}}{a^2} (\partial_x W_1 + \partial_y W_2 + \partial_z W_3) + \mathcal{O}(\epsilon^2). \quad (\text{A169})$$

Radial specialization

For a radial vector field,

$$W_i(t, \mathbf{x}) = W(t, r) \hat{r}_i, \quad r = \sqrt{x^2 + y^2 + z^2}, \quad (\text{A170})$$

we have

$$\partial_i W_i = \partial_i (W \hat{r}_i). \quad (\text{A171})$$

Using

$$\partial_i \hat{r}_i = \frac{2}{r}, \quad \hat{r}_i \partial_i W = W'(t, r), \quad (\text{A172})$$

one obtains

$$\partial_i W_i = W'(t, r) + \frac{2W(t, r)}{r}. \quad (\text{A173})$$

Therefore the linearized torsion scalar becomes

$$T_{\text{radial}} = 6H^2 - \frac{4\epsilon H}{a} \left(W'(t, r) + \frac{2W(t, r)}{r} \right) + \mathcal{O}(\epsilon^2). \quad (\text{A174})$$

Equivalently,

$$T_{\text{radial}} = \frac{6\dot{a}^2}{a^2} - \frac{4\epsilon \dot{a}}{a^2} \left(W'(t, r) + \frac{2W(t, r)}{r} \right) + \mathcal{O}(\epsilon^2). \quad (\text{A175})$$

- This equation is important for solving the exterior and interior black hole region.

Solution of Exterior Black Hole Region

For the tetrad structure to be reducible to a homogeneous scale transformation, the radial profile must satisfy

$$W'(r) = \frac{W(r)}{r}.$$

This condition is very restrictive. Indeed, it can be rewritten as

$$\frac{dW}{dr} = \frac{W}{r}.$$

Assuming $W(r) \neq 0$, we divide both sides by W :

$$\frac{1}{W} \frac{dW}{dr} = \frac{1}{r}.$$

Hence

$$\frac{d}{dr} \ln W = \frac{d}{dr} \ln r.$$

Integrating gives

$$\ln W = \ln r + \ln C,$$

where C is an integration constant. Therefore

$$W(r) = Cr.$$

Thus the homogeneous scaling condition

$$W'(r) = \frac{W(r)}{r}$$

is satisfied only by profiles of the form

$$W(r) = Cr.$$

In the exterior black-hole region, however, the Schwarzschild branch behaves as

$$W(r) \sim r^{-1/2}.$$

Let

$$W(r) = Ar^{-1/2},$$

where A is a constant. Then

$$W'(r) = -\frac{1}{2}Ar^{-3/2}.$$

On the other hand,

$$\frac{W(r)}{r} = \frac{Ar^{-1/2}}{r} = Ar^{-3/2}.$$

Therefore

$$W'(r) = -\frac{1}{2}Ar^{-3/2},$$

whereas

$$\frac{W(r)}{r} = Ar^{-3/2}.$$

Hence

$$W'(r) \neq \frac{W(r)}{r}.$$

Therefore the exterior Schwarzschild-like branch cannot be reduced to a homogeneous scale transformation. Hence, in the exterior black-hole region, does not satisfy the homogeneous scaling condition, and therefore $W(r)\hat{r}$ is not a pure diffeomorphism gauge contribution.

The Solution $W(r) = Hr$

In this appendix the radial shift field is,

$$W(r) = Hr$$

a pure diffeomorphism gauge or whether it carries genuine geometrical content.

We start from the torsion scalar used,

$$T = (\nabla \cdot W)^2 - \sum_i (\nabla W_i)^2.$$

For a radial field one writes

$$W_i = W(r)n_i,$$

where

$$n_i = \frac{x_i}{r}, \quad r = \sqrt{x^k x_k}.$$

For the homogeneous profile

$$W(r) = Hr, \quad W_i = Hr \frac{x_i}{r}.$$

Therefore

$$W_i = Hx_i.$$

Taking the spatial derivative gives

$$\partial_j W_i = \partial_j (Hx_i).$$

Assuming H is constant, this becomes

$$\partial_j W_i = H \partial_j x_i.$$

Since

$$\partial_j x_i = \delta_{ij},$$

we have

$$\partial_j W_i = H \delta_{ij}.$$

The divergence is therefore

$$\nabla \cdot W = \partial_i W_i.$$

Using

$$\partial_i W_i = H \delta_{ii},$$

and in three spatial dimensions

$$\delta_{ii} = 3,$$

we find

$$\nabla \cdot W = 3H.$$

Thus

$$(\nabla \cdot W)^2 = 9H^2.$$

Next we compute the second term,

$$\sum_i (\nabla W_i)^2 = \sum_{i,j} (\partial_j W_i)^2.$$

Using

$$\partial_j W_i = H \delta_{ij},$$

we get

$$\sum_{i,j} (\partial_j W_i)^2 = \sum_{i,j} H^2 \delta_{ij} \delta_{ij}.$$

Therefore

$$\sum_i (\nabla W_i)^2 = H^2 \sum_{i,j} \delta_{ij} \delta_{ij}.$$

In three spatial dimensions,

$$\sum_{i,j} \delta_{ij} \delta_{ij} = 3.$$

Hence

$$\sum_i (\nabla W_i)^2 = 3H^2.$$

Substituting into the torsion scalar gives

$$T = (3H)^2 - 3H^2.$$

Therefore

$$T = 9H^2 - 3H^2,$$

and finally

$$T = 6H^2.$$

Since

$$T \neq 0,$$

the configuration

$$W(r) = Hr$$

can not be interpreted as a pure diffeomorphism gauge mode.